# Research Concept Note

Modules 1 & 2 Assignment • Research Masterclass for Clinicians

## Instructions

Write a research concept note (maximum 3 pages) plus a data analysis annex (maximum 2 pages) that answers one clinical or public-health question using the course NCD–HIV cohort dataset. The dataset follows 2,500 adults in Zimbabwe from a baseline household survey (2025) to one follow-up visit (2026).

You choose one of two analysis designs:

- **Option A – Cross-sectional**: exposure and outcome both measured at baseline. You estimate prevalence.
- **Option B – Cohort**: exposure measured at baseline; new (incident) outcome measured at follow-up among people who did not have it at baseline. You estimate incidence.

Every section below must appear in your submission as its own numbered heading, in the order given. Submit by Sunday 18 October 2026, 23:59 CAT.

## The dataset

All analyses must use the course dataset provided with this brief; you do not need to register with the DHS Program. It is a simulated teaching dataset modelled on a DHS household survey with a follow-up visit, so its estimates are not official figures for Zimbabwe. Use published literature, not this dataset, to describe disease burden in your background.

| ITEM | DETAIL |
|---|---|
| Files | NCD_HIV_cohort_dataset.csv (data), NCD_HIV_cohort_data_dictionary.csv (variable names, units, codes and cut-offs), NCD_HIV_cohort_dataset_description.docx, Annex_Calculator_and_Data_Tables.xlsx (ready-made tables and calculator with dropdowns and built-in formulas for the annex), Annex_Answer_Template.docx |
| Study design | Prospective cohort. Baseline household survey July–December 2025; one follow-up visit April–September 2026 (median about 11 months) |
| Sample | 2,500 adults aged 15–64 at baseline, one row per participant, 134 variables (110 baseline, 24 follow-up) |
| Follow-up | 2,272 completed follow-up (91%), 218 lost to follow-up (9%), 10 died |
| Sampling | Two-stage cluster sample: 125 enumeration areas (cluster_id) across 10 provinces; 914 urban and 1,586 rural participants |
| Weights | sample_weight (normalised, mean = 1). Unweighted analysis is acceptable; weighted or cluster-adjusted analysis is optional. State which you use. |

| | |
|---|---|
| Content | Baseline: demographics, HIV testing and care cascade, NCD history and medicines, lifestyle, mental health scores (PHQ-9, GAD-7, SSQ-14), physical measurements and laboratory results. Follow-up (variables starting fu_): vital status, blood pressure, weight, glucose and HbA1c, PHQ-9, smoking, HIV test, ART and viral load, new TB |

## Who has data for which variables

Several variables apply only to a subgroup. Your denominator must match the subgroup.

| VARIABLES | WHO THEY APPLY TO | APPROX. N |
|---|---|---|
| Baseline demographics, HIV test result, lifestyle, mental health, BP, anthropometry | All participants | 2,500 (293 HIV-positive) |
| Baseline HIV status awareness, ART, regimen, WHO stage, TB preventive therapy | HIV-positive participants; ART variables only for those on ART | 293; 252 aware; 244 on ART |
| Baseline CD4 count, viral load, viral suppression | HIV-positive with a result | 269 |
| Baseline laboratory results | Participants who gave a blood sample | 2,352 |
| Cervical cancer screening | Women aged 25 and over | 861 |
| Follow-up measurements (fu_ variables) | Participants who completed follow-up | 2,272 |
| Follow-up glucose and HbA1c | Completed follow-up and gave a blood sample | 2,176 |
| Follow-up ART and viral load | HIV-positive at follow-up | 275 |

### Coding rules you must apply

- **N/A** means not applicable (for example, HIV care variables for HIV-negative people).
- **hypertension** and **diabetes** are measured outcomes: they combine blood pressure or glucose results with current treatment. **hypertension_diagnosed** and **diabetes_diagnosed** are self-reported awareness. Do not use them interchangeably.
- **Incident case (Option B)**: a person without the outcome at baseline (for example hypertension = No) who has it at follow-up (fu_hypertension = Yes). People who already had the outcome at baseline are not at risk and must be excluded.
- **Person-time**: use fu_followup_months (baseline interview to fu_end_date). Divide by 12 for person-years.
- **Loss to follow-up and death**: fu_status tells you who completed follow-up. Cohort analyses use those who completed follow-up.
- A viral load of 20 copies/mL means target not detected (<20).
- Use the cut-offs in the data dictionary, for example PHQ-9 ≥10 (probable depression), GAD-7 ≥10, SSQ-14 ≥8, eGFR <60, physical activity <600 MET-min/week.
- Choose an outcome with at least 30 cases (prevalent for Option A, incident for Option B). Rare outcomes include kidney disease, heart disease and stroke at baseline, and death, new TB and HIV seroconversion at follow-up.
- The dataset is for course use only. Do not share it outside the masterclass.

## Skills you will demonstrate

**MODULE 1 • WEEKS 1–6**

- Framing a public health problem epidemiologically
- Selecting and justifying an appropriate study design
- Working with population-level survey and cohort data
- Calculating and interpreting measures of disease occurrence
- Calculating and interpreting measures of effect and impact

**MODULE 2 • WEEKS 7–12**

- Classifying variables by data type and summarising them appropriately
- Applying probability and statistical distributions to study data
- Formulating and testing a statistical hypothesis
- Choosing and justifying a test for comparing means or proportions
- Calculating and interpreting a measure of association

# Required sections (maximum 3 pages)

## 1. Background and problem statement

Describe the health problem, its burden and why it warrants investigation in Zimbabwe or a similar setting. Frame it epidemiologically (person, place, time) using Week 1–2 concepts. Support burden figures with cited published sources.

## 2. Research question and objectives

State one clear, answerable research question and 2–3 specific objectives. State whether you are using Option A (cross-sectional) or Option B (cohort). The question must name:

- **Population**: who is included (for example, all adults, or HIV-positive adults only; for Option B, only those free of the outcome at baseline)
- **Exposure**: the main factor you are comparing, measured at baseline
- **Outcome**: the condition you are measuring, at baseline (Option A) or as a new case at follow-up (Option B)

Example themes (do not copy). Option A: HIV status and probable depression; urban versus rural residence and obesity. Option B: baseline obesity and incident hypertension; HIV status and new-onset depression; dolutegravir-based ART and weight gain; alcohol use and incident hypertension.

## 3. Study design and justification

State the design you are using and why it suits your question. Your study is a secondary analysis of existing data. Discuss strengths and limitations compared with the other option and with a case–control design. Cover temporality and reverse causation, and for Option B the length of follow-up and loss to follow-up (Week 6).

## 4. Data source and key variables

Cite the course dataset and describe it in 2–3 sentences (setting, years, sampling, sample size, follow-up). Then:

- State your study population and its size. For Option B, state who is at risk (free of the outcome at baseline).
- List each key variable (outcome, exposure and 2–4 other variables such as possible confounders) using its exact name from the data dictionary, and say whether it is from baseline or follow-up.
- Classify each variable by data type: nominal, ordinal, discrete or continuous (Week 8).
- State whether you will recode any variable (for example, PHQ-9 score into PHQ-9 ≥10).

A table is the recommended format for this section.

### 5. Epidemiological measures

State which measures you will calculate and what each will mean in your study (Weeks 3 and 5).

| | OPTION A – CROSS-SECTIONAL | OPTION B – COHORT |
|---|---|---|
| Occurrence | Prevalence overall and in each exposure group, with 95% CI | Cumulative incidence (risk) in each group; incidence rate per 1,000 person-years |
| Effect | Prevalence ratio, prevalence difference or prevalence odds ratio | Risk ratio, rate ratio or risk difference |
| Impact (recommended) | Population attributable fraction, interpreted with caution | Attributable risk, attributable fraction in the exposed, population attributable fraction |

Explain briefly why your measures suit your design. For example, incidence cannot be estimated from baseline data alone, and prevalence is not the same as risk.

### 6. Statistical analysis plan

Describe and justify each step of your planned analysis:

1. **Summaries (Week 8)**: counts and percentages for categorical variables; mean (SD) for approximately normal continuous variables; median (IQR) for skewed ones. Describe the baseline characteristics of your analytic sample.
2. **Distributions (Week 9)**: the distribution you assume for your outcome (for example, binomial for a proportion, Poisson for counts over person-time, normal for a mean) and how you will check it.
3. **Hypothesis (Week 10)**: the null and alternative hypotheses in words, the significance level and whether the test is one- or two-sided.
4. **Test (Week 11)**: the specific test, for example chi-square or Fisher's exact test for proportions; t-test or Mann–Whitney U for two means; ANOVA or Kruskal–Wallis for three or more groups. If you compare the same people at baseline and follow-up, use a paired test (paired t-test, Wilcoxon signed-rank or McNemar).
5. **Measure of association (Week 12)**: the measure you will report with its 95% confidence interval.
6. **Confounding, bias and survey design**: name the main confounders you would consider, how loss to follow-up could bias your result (Option B), and whether you will use the sampling weights.

## Annex. Applied statistical literacy exercise (required; maximum 2 pages; not counted in the 3-page limit)

You do not need statistical software or hand calculations for this exercise. The course calculator

(Annex_Calculator_and_Data_Tables.xlsx) already contains all the formulas. Your task is to choose a table, enter the right values, report the results and interpret them.

### Step 1. Choose a table

On the "2x2 calculator" sheet, type your name and choose any one table from the dropdown list (for example, "X1 – HIV status → Probable depression"). X tables are cross-sectional; C tables are cohort. The table does not need to match the research question in your concept note. The calculator then fills in the design, population, exposure and outcome for you.

### Step 2. Enter the values

Find your table on the "Data tables" sheet and type its four values into the yellow cells of the 2×2 table:

- **a** = exposed with the outcome
- **b** = exposed without the outcome
- **c** = unexposed with the outcome
- **d** = unexposed without the outcome

Choosing the right values is part of the assessment, so check each one against the row and column labels. The calculator then shows each result with the formula it used.

Optional for C tables: on the "Rate calculator" sheet, choose the same table and type the new cases (a and c) and the person-years.

### Step 3. Write your own interpretation

Open the "My results" sheet. In the yellow box below the summary lines, write one or two sentences in your own words explaining what your ratio and its 95% confidence interval mean, as you would to a patient or colleague. Do not copy the summary lines.

### Step 4. Copy your results into the annex

The "My results" sheet collects everything else automatically: your table, the values you entered, the measures of occurrence (prevalence or risk in each group), the measures of association (difference, ratio and odds ratio) and their 95% confidence intervals, followed by your interpretation. Copy this sheet into your annex template (select the cells, copy, and paste into Word).

### Step 5. Answer three questions

In the annex template, answer:

1. What measure of occurrence is shown (prevalence or risk), and which two groups are compared?
2. Does the 95% confidence interval include 1? What does that tell you about whether the groups truly differ, and is the difference large enough to matter clinically?
3. Name one limitation of this comparison (for example, confounding, reverse causation or loss to follow-up), and how it could affect the result.

A ratio of 1 means no difference between the groups. If the 95% confidence interval includes 1, the data do not give clear evidence of a difference.

### Step 6. Save and submit

Save the workbook as Surname_Firstname_Calculator.xlsx. Submit it with your PDF in one email (see Format and submission).

## Format and submission

| ITEM | REQUIREMENT |
|---|---|
| Length | Sections 1–6: maximum 3 pages (about 1,500 words). Annex: maximum 2 pages. References and the title page are not counted. |
| Format | PDF; 12 pt Arial or Times New Roman; 1.5 line spacing; 2.5 cm margins; numbered headings matching this brief |
| References | Vancouver style; cite the course dataset as a source |
| Due date | Sunday 18 October 2026, 23:59 CAT |
| Submission | One email to research@newlandsclinic.org.zw with the subject line "Concept Note – Surname, First name" and two attachments: (1) Surname_Firstname_ConceptNote.pdf – Sections 1–6, references and the completed annex template; (2) Surname_Firstname_Calculator.xlsx – your completed calculator workbook. |

If you need a refresher on any topic, revisit the relevant week on the Course Material page before you start writing.

# Grading rubric

Each criterion is marked at one of three levels: Excellent (80–100% of its weight), Satisfactory (50–79%) or Needs improvement (0–49%).

| CRITERION | SECTION | WEIGHT | EXCELLENT | SATISFACTORY | NEEDS IMPROVEMENT |
|---|---|---|---|---|---|
| **Background and problem statement** | 1 | **10%** | Problem framed epidemiologically, with burden supported by cited sources and clear relevance to practice. | Problem described, but framing or relevance underdeveloped. | Problem vague, uncited or not framed epidemiologically. |
| **Research question and objectives** | 2 | **10%** | Design option stated; question specific and answerable with the dataset; population, exposure and outcome defined; objectives align tightly. | Question and objectives present but broad or loosely aligned. | Question cannot be answered with the dataset, or objectives missing. |
| **Study design and justification** | 3 | **10%** | Chosen design (cross-sectional or cohort) correctly described and justified; strengths and limitations compared with alternatives, including temporality and loss to follow-up where relevant. | Design identified, but justification thin or limitations generic. | Design misidentified or not justified. |
| **Data source and key variables** | 4 | **15%** | Dataset cited; study population stated; variables named as in the dictionary, timed (baseline or follow-up) and correctly classified. | Dataset described; some classifications inconsistent or population unclear. | Key variables missing, misclassified or not in the dataset. |
| **Epidemiological measures** | 5 | **10%** | Measures of occurrence, effect and impact match the chosen design, with meaning explained in context. | Measures appropriate, but interpretation in context underdeveloped. | Measures do not match the design (for example, incidence from baseline data) or are missing. |
| **Statistical analysis plan** | 6 | **15%** | Summaries, distribution, hypotheses, test and measure of association all match the data type and design and are justified. | Plan broadly sound; one or more choices unjustified or poorly matched. | Choices missing or mismatched to the data type. |
| **Applied statistical literacy** | Annex | **20%** | Correct values a, b, c and d entered for the chosen table (checked in the submitted workbook); results copied into the annex; own interpretation of the ratio and 95% CI is correct and in plain language; questions on occurrence, the CI and a limitation answered correctly. | Values and results mostly correct, but interpretation underdeveloped or copied from the summary lines. | Wrong values entered, workbook or results missing, or interpretation missing or wrong. |

| | | | | | |
|---|---|---|---|---|---|
| **Clarity, structure and scientific writing** | Overall | **10%** | Well organised and concise; follows scientific writing conventions and the format rules. | Generally clear, with some structural or stylistic inconsistencies. | Hard to follow, or format rules not followed. |
| **Total** | | **100%** | | | |

## Checklist before you submit

- ☐ Design option (A cross-sectional or B cohort) is stated in Sections 2 and 3
- ☐ All six sections and the annex appear as numbered headings
- ☐ Sections 1–6 fit within 3 pages and the annex within 2 pages
- ☐ Exposure and outcome are variables in the dataset, named as in the data dictionary
- ☐ For Option B, people who already had the outcome at baseline are excluded from the at-risk group
- ☐ Measures and tests match your design and variable types
- ☐ Your own interpretation is written in the yellow box on the "My results" sheet, and the "My results" sheet is copied into the annex template with all three questions answered
- ☐ Sources and the dataset are cited in Vancouver style
- ☐ One email with two attachments: the PDF and the completed calculator workbook, both named correctly